

Politecnico di Milano

MSc Thesis Template — Quick Guide

Everything you need to know, plus useful commands.

This template is the official *PoliMi3i* class (School of Industrial and Information Engineering). You only edit `Thesis.tex` and `Thesis_bibliography.bib` — never the files in `Configuration_Files/`.

1 Project structure

File / folder	What it is
<code>Thesis.tex</code>	Main file — edit this
<code>Thesis_bibliography.bib</code>	Your references (BibTeX database)
<code>Configuration_Files/PoliMi3i_thesis.cls</code>	Document class — do not touch
<code>Configuration_Files/config.tex</code>	Layout settings — do not touch
<code>Images/</code>	Put your figures here

2 How to compile

The bibliography uses BibTeX, so a single pass is *not* enough. Run the full sequence:

```
pdflatex Thesis
bibtex Thesis
pdflatex Thesis
pdflatex Thesis
```

■ **NOTE** On Overleaf (this project was imported from it): just press **Recompile**. Set *Compiler* = **pdfLaTeX** and *Main document* = `Thesis.tex` in *Menu*.

■ **TIP** Locally you can use `latexmk -pdf Thesis.tex` to run the whole sequence automatically, or `tectonic Thesis.tex` (resolves citations and references in one command).

3 The first things to fill in

3.1 Title page — `\puttitle{}`

```
\puttitle{
title=My Thesis Title,
name=Name Surname,
course=Xxx Engineering - Ingegneria Xxx,
ID = 000000, % numero di matricola
advisor= Prof. Name Surname,
```

```
coadvisor={Name Surname}, % delete line if none
academicyear={2025-26},
}
```

3.2 Abstracts (both required)

PoliMi requires an abstract in **English** and one in **Italian** (*Abstract in lingua italiana*), each followed by 4–6 keywords.

3.3 Your chapters

Replace the demo chapters between `\mainmatter` and the bibliography with your own content.

4 Useful commands cheat-sheet

Command	What it does
<code>\chapter{Title}</code>	New numbered chapter
<code>\chapter*{Title}</code>	Unnumbered chapter
<code>\section{...}</code> / <code>\subsection{...}</code>	Sections (use * for no number)
<code>\input{Chapters/ch1.tex}</code>	Pull in a chapter from another file
<code>\label{ch:intro}</code>	Set a reference anchor
<code>\ref{ch:intro}</code>	Reference it (auto via cleveref)
<code>\eqref{eq:maxwell}</code>	Reference an equation
<code>\cite{key}</code>	Cite a .bib entry
<code>\includegraphics[width=...]{f}</code>	Insert an image from Images/
<code>\frontmatter</code> / <code>\mainmatter</code>	Roman (i,ii) then arabic (1,2) pages

4.1 Label naming convention

Stay consistent so cross-references read well:

Prefix	For
<code>ch:</code>	chapters
<code>sec:</code>	sections
<code>fig:</code>	figures
<code>table:</code>	tables
<code>eq:</code>	equations
<code>alg:</code>	algorithms

5 Building blocks (already demoed in the template)

5.1 Figure

```
\begin{figure}[H]
\centering
\includegraphics[width=0.6\textwidth]{myplot.png}
\caption{My caption.}
\label{fig:myplot}
\end{figure}
```

5.2 Table

```
\begin{table}[H]
\centering
\begin{tabular}{|p{3em} c c c|}
\hline
\rowcolor{bluepoli!40} % PoliMi blue header
& \textbf{col 1} & \textbf{col 2} & \textbf{col 3} \\
\hline
\hline
\textbf{row 1} & 1 & 2 & 3 \\
\hline
\end{tabular}
\caption{My caption.} \label{table:ex}
\end{table}
```

5.3 Equation

```
\begin{equation}
\label{eq:energy}
E = m c^2
\end{equation}
```

6 Bibliography

Add entries to `Thesis_bibliography.bib`, then cite with `\cite{key}`:

```
@article{einstein1905,
author = {A. Einstein},
title = {On the Electrodynamics of Moving Bodies},
journal = {Annalen der Physik},
year = {1905}
}
```

Style is `abbrvnat` (numbered, `[1]`). Change `\bibliographystyle` in `Thesis.tex` if your field prefers another.

7 Good-to-know tips

- **Split long theses:** one `.tex` file per chapter, included via `\input{}`. Easier to manage and compiles faster.

- **PoliMi blue** is predefined as `bluepoli` — use it for table headers.
- **Cell padding:** the `\T` and `\B` macros add top/bottom space inside table cells.
- **Fixed structure:** the order (Abstract EN+IT → ToC → chapters → Bibliography → Appendices → Lists → Acknowledgements) is set by PoliMi rules — keep it.
- **Image formats:** `.pdf`, `.png`, `.jpg`, `.eps` all work with pdfLaTeX. Vector (`.pdf` / `.eps`) looks sharpest.

■ **WARNING** Don't edit `PoliMi3i_thesis.cls` or `config.tex`. They control the official formatting; changing them risks a non-compliant thesis.

■ **TIP** The whole `Thesis.tex` is a **live tutorial** — every section demonstrates the feature it describes. Compile it once as-is, then replace the demo text piece by piece.